

Ethical Reinforcement Learning for Socioeconomic Fairness in Healthcare Decision Support

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Abstract. Reinforcement Learning (RL) has emerged as a powerful approach for healthcare decision support due to its ability to model sequential treatment decisions and optimize long-term clinical outcomes. However, many existing healthcare RL implementations focus predominantly on maximizing clinical rewards while implicitly assuming uniform treatment affordability and patient adherence. In real-world healthcare systems, socioeconomic constraints particularly treatment cost relative to patient economic capacity play a critical role in adherence, continuity of care, and eventual health outcomes. Ignoring these factors can result in reinforcement learning policies that disproportionately favour economically advantaged patients, thereby reinforcing socioeconomic disparities. This paper presents a technical reinforcement learning implementation that explicitly incorporates socioeconomic fairness into the policy learning process. We model healthcare decision-making as an episodic Markov Decision Process in which patient state transitions are influenced by both clinical response and cost-induced adherence dynamics. The proposed system implements a complete fairness-constrained RL pipeline using Q-learning with epsilon-greedy exploration, experience replay, and fairness-aware reward shaping. This work demonstrates the practical feasibility of embedding ethical and socioeconomic considerations directly into reinforcement learning-based healthcare decision support systems, contributing toward the development of more equitable and socially responsible healthcare AI.

Keywords: Reinforcement Learning, Ethical AI, Socioeconomic Fairness, Healthcare Decision Support, Algorithmic Bias, Fairness-Constrained Optimization

1 Introduction

Reinforcement Learning (RL) has become an increasingly important paradigm for healthcare decision support, owing to its ability to model sequential decision-making processes under uncertainty. Unlike traditional supervised learning approaches that operate on static input–output mappings, reinforcement learning explicitly captures the long-term consequences of treatment decisions by learning policies that optimize cumulative outcomes over time. This makes RL particularly well suited for healthcare applications involving chronic disease management, treatment sequencing, personalized therapy selection, and adaptive clinical decision support, where present actions influence future patient states and outcomes.

Despite this suitability, many existing healthcare reinforcement learning implementations are primarily designed to optimize clinical performance metrics such as symptom improvement, survival probability, or biomarker stabilization. These systems are commonly trained on retrospective clinical datasets and employ reward functions that focus exclusively on physiological outcomes. While clinically relevant, such formulations implicitly assume that all recommended treatments are equally affordable and that patients are uniformly able to adhere to prescribed care pathways. In real-world healthcare settings, these assumptions are rarely valid.

Socioeconomic factors including treatment cost, insurance coverage, and out-of-pocket expenditure play a decisive role in shaping patient adherence and continuity of care. Treatment strategies that are clinically optimal but financially burdensome often lead to reduced adherence, delayed intervention, or complete discontinuation, particularly among economically vulnerable populations. Reinforcement learning agents that ignore these constraints risk producing recommendations that perform well in simulation or retrospective evaluation but fail when applied to heterogeneous patient populations.

This challenge is further exacerbated by the data-driven nature of reinforcement learning. Observational healthcare datasets tend to disproportionately represent patients with consistent access to care and higher economic capacity, resulting in cleaner and more complete treatment histories. When trained on such data, reinforcement learning agents may implicitly learn policies that favor these populations, leading to feedback loops that amplify socioeconomic disparities over time. As a result, reinforcement learning systems risk reinforcing existing inequities rather than mitigating them.

This work argues that optimizing clinical outcomes alone is insufficient for responsible healthcare reinforcement learning. Instead, socioeconomic feasibility must be treated as a first-class component of the learning objective. To address this need, we present a technical reinforcement learning implementation that embeds affordability-aware fairness constraints directly into policy learning. The proposed system models healthcare decision-making as an episodic Markov Decision Process in which patient state transitions reflect both clinical response and cost-induced adherence effects. A Relative Financial Burden metric is introduced to quantify cumulative treatment cost relative to patient economic capacity, while population-level inequality is measured using the Gini coefficient and incorporated directly into the reward function.

The primary contribution of this work lies in demonstrating, through implementation, how healthcare reinforcement learning can be reframed as a joint clinical–economic optimization problem. By integrating socioeconomic fairness into the reward structure during training rather than applying post hoc bias corrections, this study provides a practical and technically grounded pathway toward ethically responsible and socially sustainable reinforcement learning–based healthcare decision support systems.

2 Application and Significance

Reinforcement learning–based decision support systems have been increasingly adopted in healthcare due to their ability to model sequential treatment decisions and optimize long-term outcomes. However, the practical effectiveness of such systems in real-world clinical environments is closely tied to socioeconomic factors, including treatment affordability, patient access to care, and cost-driven adherence behavior. Healthcare reinforcement learning systems that ignore these factors risk producing policies that perform well under idealized assumptions but fail to generalize across economically diverse patient populations. This section discusses application domains where socioeconomic-aware reinforcement learning is particularly relevant and outlines the broader significance of integrating fairness constraints directly into healthcare RL implementations.

2.1 Applications of Socioeconomically Aware Reinforcement Learning

One of the most impactful applications of reinforcement learning in healthcare is personalized treatment planning for chronic diseases, where patients require continuous and adaptive care over extended time horizons. Conditions such as diabetes, cardiovascular disease, and chronic respiratory disorders involve repeated treatment decisions that influence both immediate clinical response and long-term health trajectories. While reinforcement learning is well suited to optimizing such treatment sequences, implementations that ignore affordability may recommend high-cost interventions that patients are unable to sustain. The proposed fairness-constrained RL system addresses this limitation by modeling treatment cost and economic capacity within the environment, enabling the learned policy to balance clinical benefit with financial sustainability.

In clinical decision support systems, reinforcement learning has been applied to assist clinicians in selecting treatment options, adjusting dosages, and scheduling follow-up interventions. By explicitly incorporating cost-induced adherence dynamics and fairness-aware reward shaping, the proposed implementation supports decision-making that accounts for patient-specific economic constraints. This approach facilitates more realistic and actionable recommendations, reducing the likelihood of treatment discontinuation due to financial burden.

At the healthcare system level, reinforcement learning has been explored for optimizing resource allocation, care prioritization, and treatment planning under budget constraints. Fairness-aware reinforcement learning enables system-level policies to optimize outcomes without disproportionately favoring economically advantaged patient groups. By penalizing inequality in financial burden distribution, the proposed system provides a mechanism for promoting equitable access to healthcare resources.

across diverse socioeconomic populations.

Finally, public health and preventive care applications can benefit from affordability-aware reinforcement learning. Policies that consider economic accessibility can inform the design of intervention strategies, screening schedules, and long-term prevention programs, particularly in underserved communities where cost barriers are a primary determinant of care utilization. The implemented framework demonstrates how reinforcement learning can be extended beyond clinical efficacy to support equitable and sustainable healthcare decision-making.

2.2 Significance of the Proposed Approach

The significance of this work lies in demonstrating, through a concrete technical implementation, how socioeconomic fairness can be embedded directly into reinforcement learning-based healthcare decision support systems. By integrating affordability metrics and inequality-based penalties into the reward function during training, the proposed approach aligns algorithmic optimization with real-world feasibility and ethical considerations. This implementation-driven perspective advances healthcare reinforcement learning beyond performance-centric objectives and contributes toward the development of socially responsible, practically deployable healthcare AI systems.

From an ethical perspective, embedding fairness directly into the learning objective operationalizes principles of equity and distributive justice within reinforcement learning. This prevents the reinforcement of structural biases that may arise from historical data and feedback-driven policy learning. Practically, the approach enhances patient adherence, clinician trust, and real-world deployability of healthcare AI systems.

Overall, integrating socioeconomic fairness into reinforcement learning transforms healthcare decision support from a purely performance-driven optimization task into a human-centered, ethically grounded system capable of delivering equitable and sustainable clinical outcomes.

3 Background Study

This section reviews prior research relevant to reinforcement learning in healthcare, with particular emphasis on clinical decision support, safety-aware learning, and fairness in machine learning. The discussion highlights how existing approaches have advanced sequential decision-making in medical settings while also exposing critical limitations related to socioeconomic bias and affordability, which motivate the need for the proposed framework. A detailed study is given in Table 3.1.

3.1 Safe Reinforcement Learning via Episodic Control

Li, Zhu, and Grossklags proposed an episodic control-based framework to address one of the central challenges of safe reinforcement learning: poor sample efficiency under safety constraints. Traditional RL methods require extensive interaction with the

environment, which is infeasible in healthcare due to ethical and safety considerations. The authors mitigate this limitation by introducing state abstraction mechanisms that discretize continuous state spaces into representative clusters, enabling efficient reuse of past experiences.

A key contribution of this work is the joint optimization of reward maximization and constraint satisfaction using a Pareto-front formulation. By explicitly modeling both objectives, the method avoids collapsing safety into a secondary penalty term. Empirical evaluations demonstrate improved learning efficiency and reduced constraint violations compared to baseline approaches. However, the reliance on state abstraction introduces sensitivity to clustering quality, which may limit scalability in high-dimensional clinical settings where patient heterogeneity is pronounced.

3.2 Safe Policies for Reinforcement Learning via Primal-Dual Methods

Paternain et al. addressed safety-constrained reinforcement learning in scenarios where system dynamics are unknown, a common condition in healthcare. The authors formulated safety constraints probabilistically and introduced a primal–dual optimization framework that transforms hard constraints into a tractable Lagrangian objective. This relaxation enables gradient-based optimization without requiring explicit transition models.

The strength of this approach lies in its theoretical grounding, offering formal guarantees on constraint satisfaction under mild assumptions. Unlike heuristic penalty-based methods, the primal–dual formulation provides principled control over safety–performance trade-offs. Nevertheless, the approach assumes sufficiently expressive policy parameterizations and may face convergence challenges in highly nonconvex healthcare environments, particularly when reward and safety objectives are strongly conflicting.

3.3 Reinforcement Learning for Intelligent Healthcare Systems: A Review of Challenges, Applications, and Open Research Issues

Abdellatif et al. presented a comprehensive survey of reinforcement learning applications across intelligent healthcare architectures, spanning edge intelligence, core network optimization, and clinical decision support. The work highlights the suitability of RL for managing uncertainty, latency, and dynamic system behavior in healthcare infrastructures.

While the survey effectively maps algorithmic choices to healthcare layers, it largely emphasizes performance metrics and system efficiency, with comparatively limited focus on safety guarantees and fairness considerations. As a result, it underscores the need for deeper integration of ethical and clinical constraints into healthcare RL frameworks rather than treating them as secondary design concerns.

3.4 Fair Machine Learning in Healthcare: A Survey

Feng et al. provided a foundational analysis of fairness in healthcare machine learning grounded in principles of distributive justice. The study categorizes fairness challenges

into equal allocation and equal performance, offering a structured lens for evaluating algorithmic equity. Importantly, the authors trace sources of bias across the entire ML lifecycle, from data collection to deployment.

Although the survey offers extensive coverage of fairness metrics and mitigation strategies, its primary focus is on supervised learning paradigms. Extending these fairness concepts to reinforcement learning where decisions affect future states and data distributions introduces additional complexity that remains insufficiently explored. This gap is particularly critical for healthcare applications involving long-term treatment policies.

3.5 A Reinforcement Learning Model for Optimal Treatment Strategies in Intensive Care

Drudi et al. proposed reinforcement learning-based frameworks for optimizing treatment strategies in intensive care units, with particular emphasis on cardiorespiratory dynamics and physiological feature importance. Their work demonstrates that RL models can learn complex, temporally dependent treatment policies from ICU data, identifying clinically meaningful state variables that influence therapeutic decisions.

A notable contribution of this line of research is the incorporation of feature importance analysis, which partially addresses clinician concerns regarding interpretability. However, these studies primarily rely on retrospective observational datasets and counterfactual evaluations. As a result, the causal validity and generalizability of the learned policies across diverse ICU populations remain open questions, underscoring the need for prospective clinical validation.

3.6 Warfarin Dose Management Using Offline Deep Reinforcement Learning

Ji et al. investigated the application of offline deep reinforcement learning for personalized warfarin dose management using historical electronic health records. By restricting learning to pre-collected data, their approach eliminates the need for online exploration, which is infeasible in high-risk anticoagulation therapy.

The results indicate that offline RL can achieve dosing strategies comparable to, or in some cases exceeding, expert clinician performance while maintaining patients within therapeutic ranges. Despite these promising outcomes, the approach remains sensitive to dataset bias and distributional shift. Performance degradation may occur when deployment populations differ substantially from training cohorts, highlighting the importance of robust off-policy evaluation and uncertainty quantification.

3.7 Adaptive Decision Transformer for Dynamic Treatment Regimes in Sepsis

Jeon et al. introduced Adaptive Decision Transformer (ADT) architectures tailored for dynamic treatment regimes in sepsis management. By leveraging transformer-based attention mechanisms, ADTs model clinical decision-making as a sequence prediction problem, enabling the identification of prior observations that most strongly influence current treatment actions.

This architecture improves interpretability relative to conventional deep RL methods and aligns well with clinician reasoning processes. However, transformer-based approaches typically lack explicit safety guarantees and may struggle with long-horizon credit assignment. Integrating attention-based models with formal safety constraints and fairness-aware objectives remains an important open research direction.

3.8 Exploring Bias and Prediction Metrics to Characterize Fairness of Machine Learning for Equity-Centered Public Health Decision-Making

Raza et al. examined bias and fairness in public health machine learning systems, emphasizing the limitations of aggregate performance metrics in capturing inequities across demographic subgroups. Their work highlights how conventional evaluation practices can mask disparities in sensitivity, specificity, and predictive utility among vulnerable populations.

While the study provides valuable insights into equity-centered evaluation, its focus is primarily on static predictive models. Extending these fairness considerations to reinforcement learning systems introduces additional challenges, as policy decisions influence future health outcomes and data distributions. This limitation underscores the need for fairness metrics specifically designed for sequential and adaptive decision-making.

3.9 Multi-Objective Reinforcement Learning Based Healthcare Expansion Planning Considering Pandemic Events

Shuvo et al. applied multi-objective reinforcement learning to healthcare infrastructure expansion and capacity planning during pandemic scenarios. Their framework balances competing objectives such as infection control, economic cost, infrastructure development, and equitable access to care.

Although effective at the system-planning level, these approaches operate at a high level of abstraction and do not directly address patient-level safety or clinical interpretability. Bridging the gap between strategic planning models and individualized clinical decision support remains a non-trivial limitation for healthcare RL research.

3.10 AoI-Aware Resource Management for Smart Health Via Deep Reinforcement Learning

Wu et al. proposed deep reinforcement learning approaches for Age-of-Information (AoI)-aware resource management in smart healthcare systems. Their work optimizes monitoring and communication policies to balance information freshness against computational and communication costs, ensuring timely detection of patient deterioration.

3.11 Reinforcement Learning-Based Integrated Risk-Aware Dynamic Treatment Strategy for Consumer-Centric Next-Gen Healthcare

In parallel, Nimma et al. developed risk-aware reinforcement learning frameworks that explicitly incorporate patient risk into treatment optimization objectives. While both approaches align algorithmic behavior with clinical priorities, risk modeling is often simplified, and interactions between risk, fairness, and long-term outcomes are insufficiently explored.

3.12 Pruning the Way to Reliable Policies: A Multi-Objective Deep Q-Learning Approach to Critical Care

Shirali et al. introduced a multi-objective deep Q-learning framework with policy pruning to improve reliability and interpretability in critical care decision-making. By identifying and removing unstable or spurious learned behaviors, the method enhances policy robustness and clinical trust.

Despite these advances, scalability to high-dimensional state spaces and robustness under clinical variability remain unresolved. Additionally, balancing multiple competing objectives under strict safety constraints continues to pose significant algorithmic challenges.

Table 3.1 presents a comprehensive summary of all the related works which has already been done on Reinforcement Learning in Healthcare and Financial Decision Making.

Table 3.1. Summary of Related Work in Reinforcement Learning for Healthcare

S. No	Author Name, Journal & Year	Title of the Paper	Observations	Technology Used
1	Li, Zhu, and Grossklags, <i>Journal of Artificial Intelligence Research</i> , 2018	Safe Reinforcement Learning via Episodic Control	Improves sample efficiency under safety constraints using episodic memory and state abstraction; sensitive to clustering quality in complex healthcare settings	Episodic Control, Safe Reinforcement Learning, State Abstraction
2	Paternain et al., <i>IEEE Transactions on Automatic Control</i> , 2019	Safe Policies for Reinforcement Learning via Primal-Dual Methods	Provides theoretical guarantees for safety-constrained RL using Lagrangian relaxation; limited scalability in highly nonconvex environments	Primal–Dual Optimization, Constrained Reinforcement Learning
3	Abdellatif et al., <i>IEEE Communications Surveys & Tutorials</i> , 2021	Reinforcement Learning for Intelligent Healthcare Systems: A Review of Challenges, Applications, and Open Research Issues	Comprehensive survey of RL applications in healthcare; limited emphasis on fairness and ethical constraints	Reinforcement Learning, Intelligent Healthcare Systems

4	Feng et al., <i>ACM Computing Surveys</i> , 2022	Fair Machine Learning in Healthcare: A Survey	Identifies bias sources across the ML lifecycle; focuses mainly on supervised learning, not sequential RL settings	Fair Machine Learning, Bias Mitigation Techniques
5	Drudi et al., <i>Nature Scientific Reports</i> , 2021	A Reinforcement Learning Model for Optimal Treatment Strategies in Intensive Care	Demonstrates RL-based treatment optimization in ICUs; relies on retrospective data and counterfactual evaluation	Reinforcement Learning, ICU Decision Support
6	Ji et al., <i>Proceedings of the AAAI Conference on Artificial Intelligence</i> , 2020	Warfarin Dose Management Using Offline Deep Reinforcement Learning	Shows offline RL can match clinician-level dosing performance; sensitive to dataset bias and distributional shift	Offline Deep Reinforcement Learning
7	Jeon et al., <i>NeurIPS</i> , 2023	Adaptive Decision Transformer for Dynamic Treatment Regimes in Sepsis	Uses transformer architectures for sequential clinical decisions; lacks explicit safety and fairness constraints	Decision Transformers, Sequence Modeling
8	Raza et al., <i>Journal of Biomedical Informatics</i> , 2022	Exploring Bias and Prediction Metrics to Characterize Fairness of Machine Learning for Equity-Centered Public Health Decision-Making	Highlights limitations of aggregate metrics in fairness evaluation; focuses on static predictive models	Fairness Metrics, Public Health ML
9	Shuvo et al., <i>IEEE Systems Journal</i> , 2021	Multi-Objective Reinforcement Learning Based Healthcare Expansion Planning Considering Pandemic Events	Balances cost, capacity, and access during pandemics; operates at system-planning rather than patient level	Multi-Objective Reinforcement Learning
10	Wu et al., <i>IEEE Internet of Things Journal</i> , 2020	AoI-Aware Resource Management for Smart Health via Deep Reinforcement Learning	Optimizes data freshness and communication efficiency in smart health systems	Deep Reinforcement Learning, Age-of-Information
11	Nimma et al., <i>IEEE Journal of Biomedical and Health Informatics</i> , 2022	Reinforcement Learning-Based Integrated Risk-Aware Dynamic Treatment Strategy for Consumer-Centric Next-Gen Healthcare	Incorporates patient risk into RL objectives; fairness and affordability interactions remain unexplored	Risk-Aware Reinforcement Learning
12	Shirali et al., <i>Artificial Intelligence in Medicine</i> , 2021	Pruning the Way to Reliable Policies: A Multi-Objective Deep Q-Learning Approach to Critical Care	Improves robustness and interpretability using policy pruning; scalability challenges persist	Multi-Objective Deep Q-Learning

4 Research Gaps

Although reinforcement learning has demonstrated strong potential for healthcare decision support, a closer examination of existing implementations reveals several unresolved gaps that limit their ethical robustness, practical feasibility, and equitable impact. These gaps are most evident at the intersection of sequential decision-making, treatment affordability, and fairness-aware policy optimization. This section identifies the specific research gaps that are directly addressed by the implemented fairness-constrained reinforcement learning system presented in this work.

4.1 Lack of Affordability Modeling in RL Environments

Most existing healthcare reinforcement learning implementations define patient state transitions exclusively in terms of clinical variables such as disease severity, physiological response, or treatment efficacy. Treatment cost and patient economic capacity are typically excluded from the environment dynamics.

Gap: Current RL environments fail to model treatment affordability as a factor influencing patient state evolution, leading to policies that ignore cost-induced effects on adherence and long-term outcomes.

4.2 Absence of Fairness Constraints During Policy Learning

Fairness considerations in healthcare machine learning are often evaluated after training, rather than enforced during policy optimization. In reinforcement learning, this omission is particularly problematic because learned policies shape future data distributions through feedback loops.

Gap: Existing healthcare RL implementations do not enforce fairness constraints during training, allowing socioeconomic bias to emerge and amplify over time.

4.3 No Explicit Population-Level Inequality Metric in RL Reward Design

While some healthcare RL systems incorporate cost-sensitive rewards, these formulations typically penalize absolute cost without accounting for how financial burden is distributed across patients.

Gap: There is no explicit mechanism in existing healthcare RL implementations to quantify and penalize inequality in financial burden across a simulated patient population.

4.3 Oversimplified Assumptions of Treatment Adherence

Many reinforcement learning models implicitly assume full patient adherence to prescribed treatments, treating deviations as stochastic noise or clinical failure.

Gap: Cost-induced non-adherence is not modeled explicitly, resulting in policies that may perform well in simulation but fail under realistic economic constraints.

4.4 Bias Amplification from Observational Data Assumptions

Healthcare reinforcement learning systems are often trained or evaluated under assumptions derived from observational datasets that overrepresent economically advantaged patients with consistent care histories.

Gap: Existing RL implementations do not mitigate bias amplification arising from training dynamics that favor populations with cleaner or more complete data.

4.5 Disconnect Between Ethical Objectives and Algorithmic Design

Ethical principles such as equity and distributive justice are frequently discussed at a conceptual or policy level but are rarely operationalized within reinforcement learning algorithms.

Gap: There remains a disconnect between ethical intent and technical implementation in healthcare reinforcement learning systems.

4.6 Summary of Addressed Gaps

In summary, existing healthcare reinforcement learning implementations commonly lack:

- Explicit modeling of treatment affordability within environment dynamics
- Enforcement of fairness constraints during policy learning
- Quantitative population-level inequality metrics in reward design
- Explicit modeling of cost-induced non-adherence
- Mechanisms to limit socioeconomic bias amplification during training

Table 4.1 presents a complete idea on all the found research gaps and the impact they have on the system. The implemented fairness-constrained reinforcement learning system directly addresses these gaps by embedding affordability-aware state transitions and inequality-based reward penalties into the learning process, enabling ethically grounded, economically realistic, and clinically meaningful healthcare decision support.

Table 4.1: Summary of Research Gaps and Impact on the system

Research Gap	Technical Description	Impact on System
Lack of Affordability Modeling in RL Environments	Most healthcare RL environments define patient state transitions only using clinical variables such as disease severity or treatment efficacy. Treatment cost and patient economic capacity are excluded from environment dynamics.	Policies ignore cost-induced effects on adherence and long-term outcomes, resulting in clinically optimal but economically infeasible treatment recommendations.
Absence of Fairness	Fairness is typically evaluated after training rather than enforced during	Socioeconomic bias can emerge and amplify over time,

Constraints During Policy Learning	reinforcement learning optimization, despite RL policies influencing future data distributions through feedback loops.	leading to inequitable treatment policies that favor economically advantaged patients.
No Explicit Population-Level Inequality Metric in Reward Design	Cost-sensitive reward formulations penalize absolute cost but do not measure how financial burden is distributed across patients within a population.	The system cannot detect or penalize unequal financial burden, allowing policies that disproportionately disadvantage low-income patients.
Oversimplified Assumptions of Treatment Adherence	Many RL models assume full patient adherence and treat deviations as stochastic noise or clinical failure, without modeling affordability-driven non-adherence.	Learned policies perform well in simulation but fail under realistic economic constraints where cost directly affects adherence behavior.
Bias Amplification from Observational Data Assumptions	RL systems are often trained or evaluated using assumptions derived from observational datasets that overrepresent patients with consistent access to care and cleaner histories.	Training dynamics favor economically advantaged populations, reinforcing and amplifying existing socioeconomic disparities.
Disconnect Between Ethical Objectives and Algorithmic Design	Ethical principles such as equity and distributive justice are discussed conceptually but are not translated into measurable objectives within RL algorithms.	Ethical goals remain external to optimization, preventing systematic enforcement of fairness during policy learning.

5 Problem Statement

Healthcare treatment planning can be modeled as a sequential decision-making process in which clinical actions influence both immediate patient response and long-term health trajectories. Reinforcement learning provides a natural framework for learning such treatment policies by optimizing cumulative reward over time. However, conventional healthcare reinforcement learning implementations remain incomplete, as they typically optimize clinical effectiveness while neglecting socioeconomic feasibility and equity, both of which strongly influence real-world treatment adherence and outcomes.

5.1 Conventional Reinforcement Learning Formulation

Let the healthcare decision process be represented as a Markov Decision Process (MDP) defined by the tuple (1)

$$M = \langle S, A, P, R, \gamma \rangle \quad (1)$$

where:

S denotes the patient state space (clinical indicators, history),
 A represents available treatment actions,
 $P(s' | s, a)$ is the state transition probability,
 $R(s, a)$ is the reward function capturing clinical outcomes,
 $\gamma \in (0,1)$ is the discount factor.

Unlike conventional formulations, the implemented environment explicitly incorporates treatment cost and patient economic capacity into state transitions, allowing financial burden to influence adherence behavior and future patient states. This formulation relaxes the unrealistic assumption of uniform affordability and full treatment adherence commonly adopted in prior work.

5.2 Problem of Socioeconomic Bias in Policy Learning

Patients differ substantially in economic capacity, insurance coverage, and tolerance for out-of-pocket expenses. Let $C(s, a)$ denote the financial cost associated with a treatment action and I_i represent the economic capacity of patient i . Reinforcement learning policies that maximize clinical reward without accounting for affordability may impose disproportionate financial burden on economically vulnerable patients.

When trained in environments that ignore these factors, reinforcement learning agents tend to favor high-cost treatments that perform well clinically but are infeasible for large segments of the population. This leads to systematic socioeconomic bias in policy learning, particularly when observational assumptions implicitly reflect economically advantaged patient cohorts.

Formally, the expected return optimized by conventional RL as in (2),

$$\max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^T \gamma^t R(s_t, a_t) \right] \quad (2)$$

which does not penalize unequal distribution of treatment burden across patients.

5.3 Affordability Modeling via Relative Financial Burden

To explicitly capture treatment affordability, the implemented system defines **Relative Financial Burden (RFB)** for each patient as in (3).

$$RFB_i = \frac{C_i}{I_i} \quad (3)$$

where C_i denotes the cumulative treatment cost incurred by patient i and I_i represents their estimated economic capacity. Let G denote the distribution of RFB across the patient population. Socioeconomic inequity can be quantified using the **Gini coefficient** $G(G)$, which measures inequality in financial burden distribution.

5.4 Fairness-Constrained Reinforcement Learning Objective

The core problem addressed in this work is to learn a policy π^* that achieves high clinical effectiveness while minimizing socioeconomic inequality. This is formulated as a constrained optimization problem as in (4).

$$\max \mathbb{E}_{\pi} \left[\sum_{t=0}^T \gamma^t (R_{clinical}(s_t, a_t) - \lambda * G) \right] \quad (4)$$

where G denotes the Gini coefficient of the Relative Financial Burden distribution and λ controls the trade-off between clinical performance and socioeconomic fairness.

This formulation ensures that fairness considerations directly influence action selection and policy updates during training rather than being applied as post hoc evaluation criteria.

5.5 Statement of the Implemented Research Problem

The research problem addressed by this implementation is defined as follows:

How can a reinforcement learning-based healthcare decision support system be implemented such that treatment policies optimize long-term clinical outcomes while explicitly accounting for treatment affordability and limiting socioeconomic inequality during policy learning?

This work addresses the problem by embedding affordability-aware state dynamics and inequality-based reward penalties directly into a reinforcement learning training pipeline. By operationalizing ethical objectives within the algorithmic design, the implemented system aligns clinical optimization with principles of equity, sustainability, and patient-centered healthcare. Figure 5.1 represents the system architecture of the proposed model

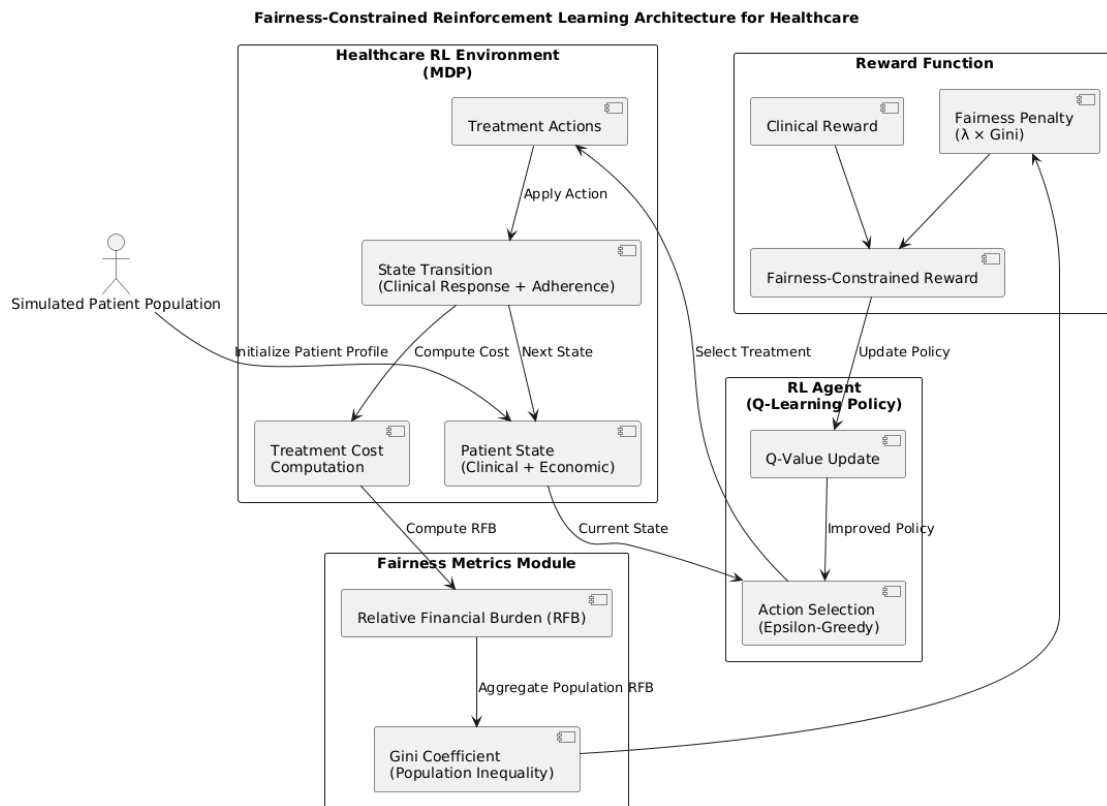


Fig 5.1: System Architecture of the proposed model